



# FERIT

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**Embedded Systems**

## **Peugeot Infotainment Project**

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**1. rujna 2025.**

**Academic year 2024/2025**

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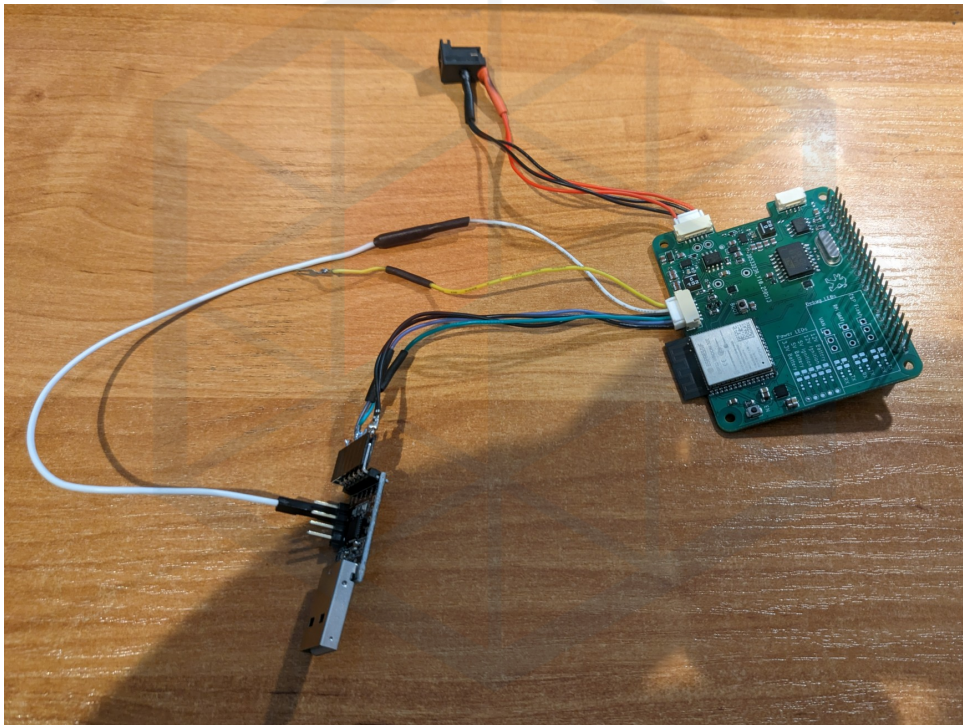
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# 1 The project

## 1.1 Intro

The goal of this project was to create a gateway module for an infotainment system that will be used in a Peugeot 206. The system itself has two major components: the gateway module and the infotainment computer. The gateway module is a custom board based on an ESP32 Wroom 32E module. The board handles power, communication with the car and communication with the infotainment computer. The infotainment computer is a Raspberry Pi 4 with a DSI touchscreen display and a custom Linux distribution. The computer handles the display and audio.

Ovaj project je proširenje postojećeg završnog rada ‘VAN protokol i njegova primjena u automoblima’ [5].



Slika 1: The gateway module setup for development

## 1.2 Main Challenges

This project has two, maybe three main challenges to consider. The obvious one is car communication. The Peugeot 206 doesn't use standard CAN bus communication, but rather something called VAN, see chapter 2. This communication standard is obsolete today, as it was only used in Peugeot and Citroën cars for a limited time. Because of that, almost no hardware exists that can understand the protocol, and about as much documentation, most of it in French. Therefore, we must create our own parser for receiving VAN frames, and figure out how to send them as well.

The second challenge would be power. We have components requiring both 5 Volts and 3.3 Volts, as well as two power rails: a permanent source and a switched source. The other problem is **very** noisy and, sometimes, unstable input power, so cheap DC-DC converters will not cut it.

The last challenge, and maybe the most underestimated one, is writing a robust firmware for the gateway module itself. The module doesn't just need to parse the VAN frames, it needs to manage its

own state, figure out the car state, handle the infotainment computer's power input and communication, handle sleep states and manage its own low power state, and, most importantly, not crash.



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## 2 VAN Bus

VAN (Vehicle Area Network) is a serial communication bus developed by the PSA group (Peugeot, Citroën) and Renault as an alternative to the CAN bus. It's defined in the ISO standard **ISO 11519-3**. It utilizes the same physical layer as the CAN bus (a differential pair) and identical bus arbitration method, but it differs in the way the data is packaged. Almost all of the information here was sourced from the Atmel TSS463C datasheet [3].

Like CAN, the VAN protocol uses identifiers to differentiate between messages and perform bus arbitration. Additionally, it also has the ability for so called "In-frame" responses, bigger message sizes (28 bytes instead of 8 bytes in CAN) and the ability to directly communicate with another device.

Data on the VAN network is packed into frames. Frames consist of multiple parts: Start of frame (**SOF**), message identifier (**IDEN**), command (**COM**), message data (**DATA**), frame checksum (**FCS**), end of data signal (**EOD**), acknowledge signal (**ACK**) and the end of frame signal (**EOF**).

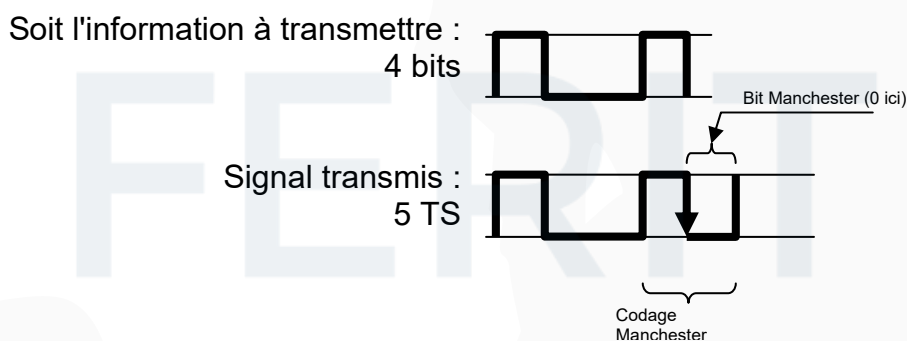
| SOF   | IDEN            | Command<br>(4 bit, 5 TS) |     |     |     | DATA max. | FCS    | EOD  | ACK  | EOF  |
|-------|-----------------|--------------------------|-----|-----|-----|-----------|--------|------|------|------|
| 10 TS | 12 bit<br>15 TS | EXT                      | RAK | R/W | RTR | 28 byte   | 15 bit | 2 TS | 2 TS | 4 TS |
|       |                 |                          |     |     |     | 280 TS    | 18 TS  |      |      |      |

Tablica 1: VAN frame format

### 2.1 Enhanced Manchester code

All VAN frames are coded using Enhanced Manchester coding (e-Manchester). With e-Manchester, after sending 3 NRZ bits, the fourth bit is sent as a Manchester bit. Sending data this way ensures easier synchronization on the bus. This is a more predictable method of bitstuffing as opposed to the CAN bus method.

In Table 1 we can see the number of bits for each part of the frame, as well as the number of Time Slots (TS). Since a Manchester bit takes two time slots to send, 4 bits in a row encoded with e-Manchester takes  $3 + 2$  Time slots to send.

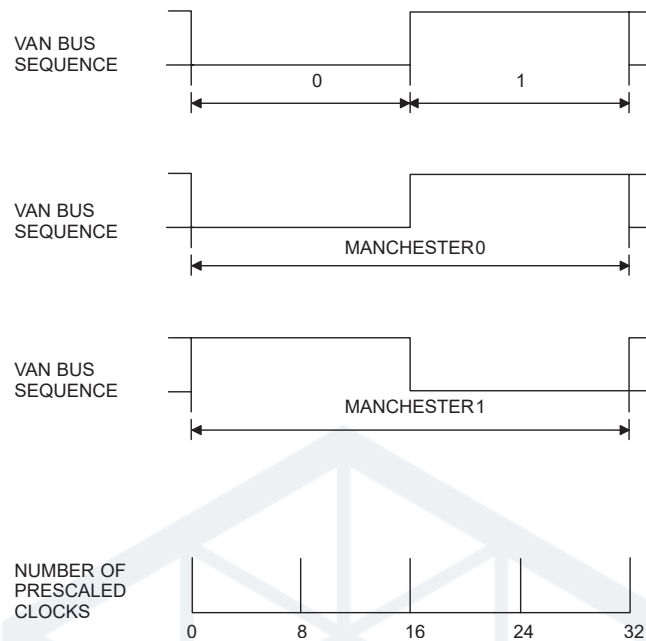


Slika 2: e-Manchester encoding [1]

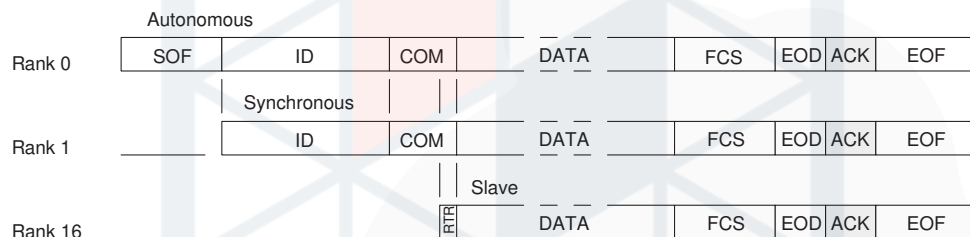
### 2.2 Messages

A device/module connected to the network can be one of three types:

- The Autonomous module. A bus master. It can send **SOF** sequences, initiate data transfers and receive messages.



Slika 3: NRZ and Manchester bits. [3]



Slika 4: Module types

- The Synchronous access module. It cannot send **SOF** sequences, but can initiate data transfers and receive messages.
- The Slave module. It can only send messages using in-frame responses and can also receive messages.

The VAN protocol supports multiple message types. They are differentiated using the identifier (**IDEN**) and command (**COM**) fields. The identifier is 12 bits long and is followed by the 4 command bits. Due to the way the bus arbitration works, a smaller identifier has priority on the bus. While the identifier value is completely up to the user, the command bits are strictly defined and determine the message type that will be sent on the bus:

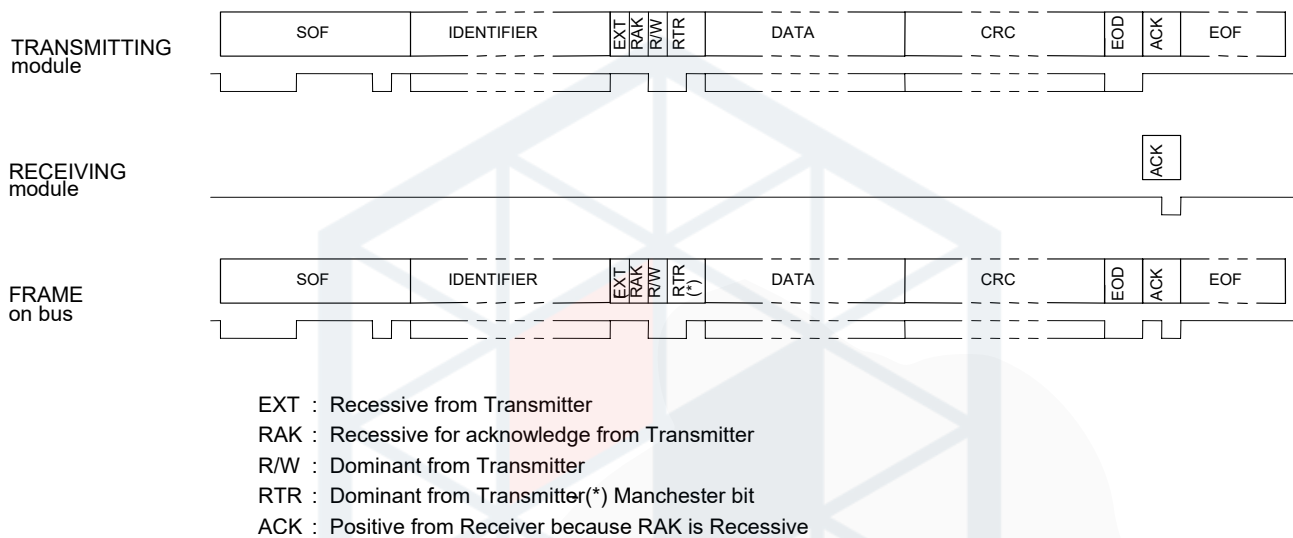
- EXT - always 1
- RAK - „Request Acknowledge“. If 1, the recipient must acknowledge the message reception. If 0, the ACK signal must not be sent.
- R/W - „Read/Write“. Signals the data direction. If 0, then it's a 'write' message, a device is sending data intended for another device. If 1, then it's a 'read' message, and it implies a request for data and expects an answer.
- RTR - „Remote Transmission Request“. If 0, the message contains data. If 1, the message contains no data and an in-frame response request is implied. The combination R/W = 0 and RTR = 1 is forbidden on the bus.

### 2.2.1 Message types

Here we will list frame types relevant to this project. There are more, but we only use these.  
Reminder: Dominant = 0, Recessive = 1.

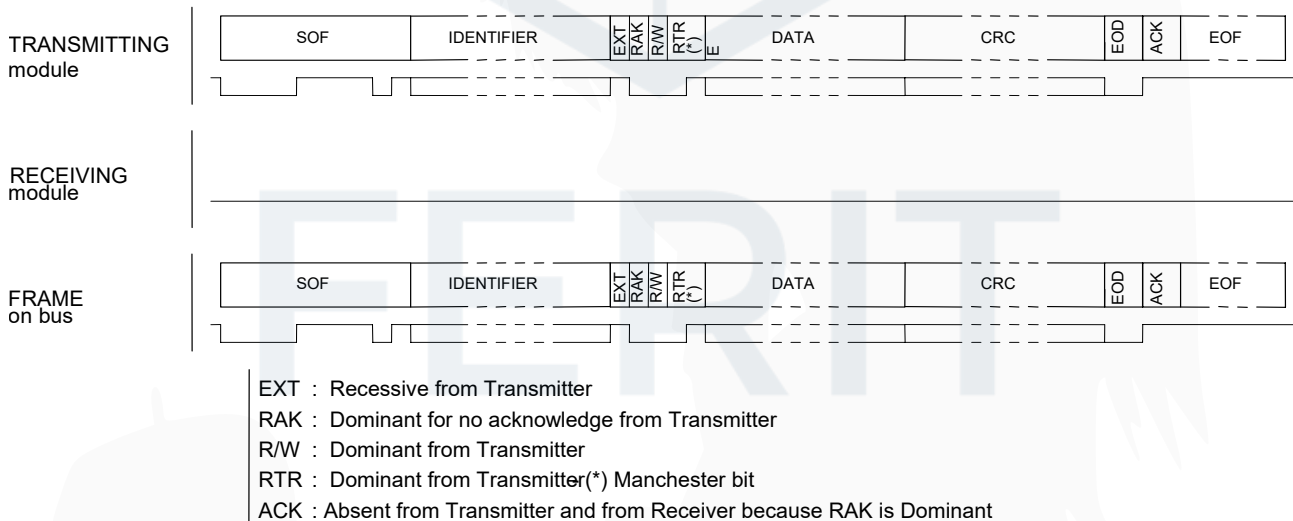
**Normal frame** The simplest frame type. One module sends an entire frame with data, and may or may not request an ACK. In the case no ACK is requested, the message is treated as a broadcast message.

With acknowledgment



Slika 5: Normal frame transimssion with ACK

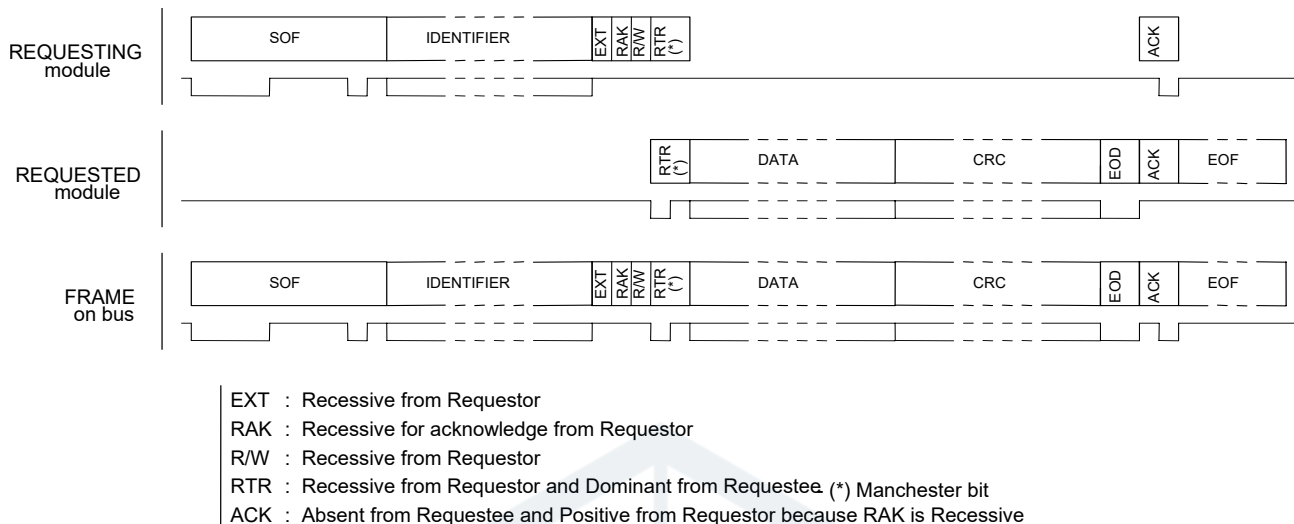
Without acknowledgment



Slika 6: Normal frame transimssion without ACK

**Reply Request Frame with Immediate Reply** This is the only frame in which a slave module can transmit data. The only difference on the bus is the R/W bit changed state compared to the normal frame. The bus master generates the **SOF**, the initiator generates the identifier and first 3 bits of the command, and the ACK signal. The rest (RTR, data and checksum) are generated by the replying module.

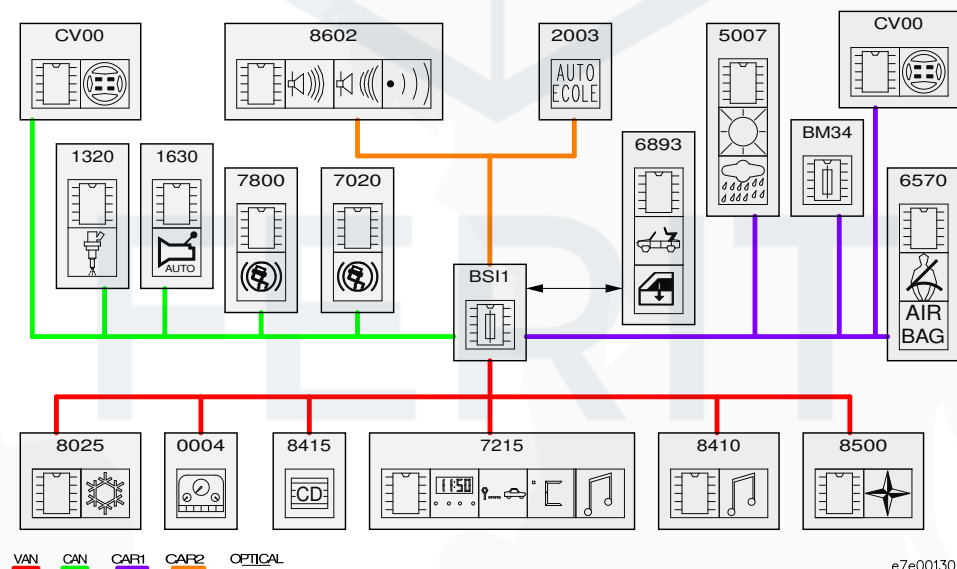




Slika 7: Reply Request Frame with Immediate Reply

## 2.3 VAN network

We can connect multiple modules together to create a network. We can also connect multiple networks together and use a gateway module to route data between networks. In the Peugeot 206 we have 1 CAN network *CAN Engine* and 3 VAN networks: *VAN Comfort*, *VAN Body 1* and *VAN Body 2*. We can see the layout in figure 8. From there we also see the gateway module, the BSI. This module is present in all PSA group cars. This particular BSI is from the AEE2001 generation, the only generation to use the VAN bus. Other generations, the AEE2004 and AEE2010 both exclusively use CAN. Our infotainment system will be connected in place of the CD changer, module number 8415 on figure 8 on the *VAN Comfort* bus.



Slika 8: VAN and CAN Network layout in the Peugeot 206 [2]

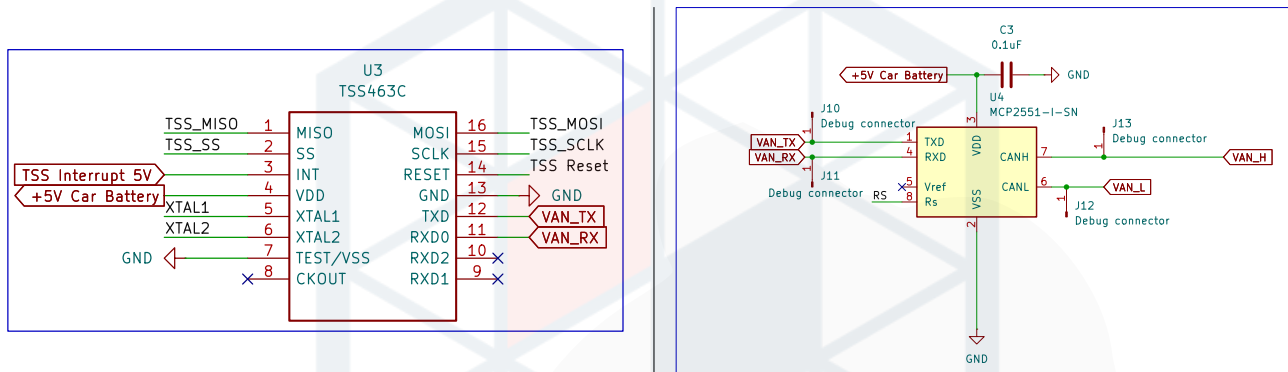


### 3 Gateway module hardware

The gateway module is designed as a Raspberry Pi HAT board. It has to contain necessary hardware for power regulation and car communication. It also has to expose all IO for connecting external devices such as an ESP32 programmer and a power cable for the Raspberry Pi display.

#### 3.1 Communication hardware

In section 2 we have established that the VAN bus is a differential protocol much like the CAN bus and, as a result, we can use readily available CAN transceiver hardware. However, we still need hardware that can generate valid VAN frames. For that we use the Atmel TSS463C VAN driver chip. It is the only VAN driver chip that exists. For the CAN transceiver we use the MCP2551 because it was proven to work with early prototypes and, with its 5 volt logic, integrates nicely with the TSS.



Slika 9: TSS463C and MCP2551 in the schematic

##### 3.1.1 TSS463C

The TSS463C is the VAN driver chip we use for sending VAN frames to the car. It is the only standalone VAN driver chip that exists, as all other modules inside the car have a VAN driver built into their SOCs. The TSS uses SPI to communicate with the ESP32.

#### 3.2 Voltage levels

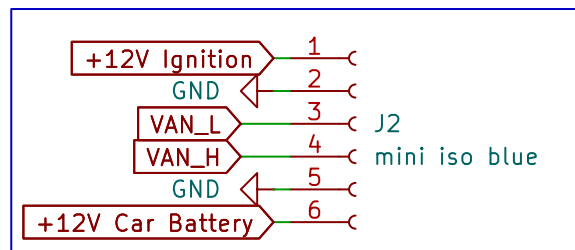
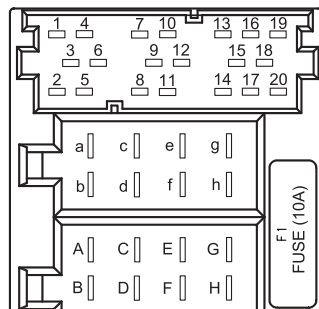
The TSS463C and the MCP2551 both use 5V logic, while all other components on the board, including the ESP32, use 3.3V logic. For that we use a pair of NTS0104 level shifters. One of them is used for the TSS SPI communication with the ESP, while the other is used for connecting the VAN RX line from the MCP to the ESP32 for reading VAN frames, and the TSS interrupt pin.

#### 3.3 Connecting to the car

We are connecting to the car in place of the CD Changer unit. The CD Changer connects to the original RD3 Headunit in the C3 (pins 13 to 20) portion of the ISO 10487 radio connector. See Figure 10 for the pinout.

Pins 13 to 17 for data and power are connected to the gateway module with a JST GH 6 pin connector, while pins 18 to 20 are wired to a 3.5mm audio cable and connected to the Raspberry Pi. Since the naming convention is different between Figures 10 and 11, see

| No. | Description   |
|-----|---------------|
| 1   | N.C.          |
| 2   | N.C.          |
| 3   | N.C.          |
| 4   | N.C.          |
| 5   | N.C.          |
| 6   | N.C.          |
| 7   | VOICE SYN     |
| 8   | VOICE SYN-GND |
| 9   | N.C.          |
| 10  | N.C.          |
| 11  | AUX(+)        |
| 12  | AUX(-)        |
| 13  | VAN DATA      |
| 14  | VAN DATA/     |
| 15  | CD A/C GND    |
| 16  | CD A/C B/U    |
| 17  | +VAN          |
| 18  | CD A/C S-GND  |
| 19  | CD A/C L      |
| 20  | CD A/C R      |



Slika 11: JST Connector in the schematic

Slika 10: ISO connector pinout [4]

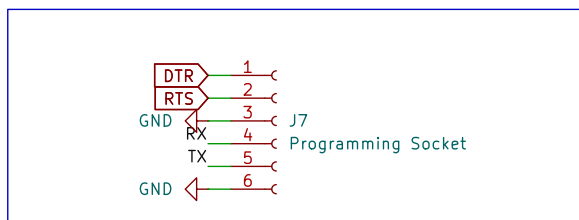
|            |   |                  |
|------------|---|------------------|
| VAN DATA   | ⇔ | VAN_L            |
| VAN DATA/  | ⇔ | VAN_H            |
| CD A/C GND | ⇔ | GND              |
| CD A/C B/U | ⇔ | +12V Car Battery |
| +VAN       | ⇔ | +12V Ignition    |

Tablica 2: ISO to JST connection legend

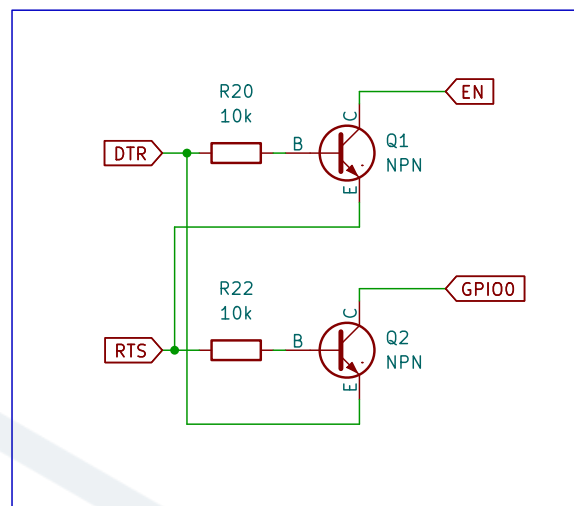
### 3.4 ESP32 Flashing

Flashing firmware on the ESP32 is done using UART. In order to flash the firmware, the ESP must first be put into flashing mode. This can be done by pulling pins GPIO00 and GPIO02 low when powering on the board. This can be achieved in two ways on our board: Manually by pressing the BOOT momentary button while resetting the board with the EN button, or by connecting the DTR and RTS signals of the UART adapter to the appropriate pins on the JST programming connector. They are connected internally to the EN and GPIO00 inputs and the ESP programmer (esptool.py) supports resetting and flashing the board using those signals.

The UART programming interface is exposed on another JST GH 6 pin connector as well as the DTR and RTS signals.



Slika 12: Programming header pinout



Slika 13: DTR and RTS signals

### 3.5 Power

For power, we have two 12V voltage rails to work with. The first one is always live (+12V Car Battery), and the second one is switched on only when the car is awake (+12V Ignition). The second rail is incorrectly labeled as it doesn't exactly follow the ignition key state, but the car's awake state. The car will wake up when an action happens (locks lock/unlock, doors open/close, key gets put in the ACC position, etc.) and will go to sleep after some time if nothing is happening and the key is not in the ACC position.

As we have two rails, we have two separate voltage regulators (MP8759GD1) tasked with turning the 12 V from the input into 5 V at the output. The voltage regulator on the +12V battery rail is used to power the ESP32, TSS, and MCP chips, as they need to be able to be active even when the car is in sleep mode. However, as the ESP32 requires 3.3 V, we had to expand the design with a single 3.3V LDO voltage regulator.

The other 5 V voltage regulator is tasked with supplying the RPi and the peripheral LCD with power. As they are not used for communicating with the car, there is no need for them to have a constant power supply.

Apart from the voltage regulator ICs, on the schematic there is an outstanding number of filtering capacitors. They are required to provide a stable voltage source for the other ICs, as some of them are quite sensitive to noise and could start behaving in unexpected ways. Another, often overlooked component when designing power supplies is the TVS diode, which is connected to the regulator input in a reverse polarized configuration, ensuring a maximum voltage of 12 V.

### 3.6 Raspberry Pi

The RPi is a slave to the ESP32 in the sense that it does not handle any communication with the car or any other system-critical matter. Its job is to render the telemetry data received from the ESP32 over UART on the LCD and handle the *entertainment* part of the infotainment system. As a noncritical part of the system, there is no need for uninterrupted power access; hence, it is powered only when the car is awake according to table 3.

### 3.7 Misc

**RTC** As the existing car display doesn't report the time on the bus, an additional RTC was needed. The selected RTC is DS3231MZ. It's connected to the Raspberry Pi over I2C. It is readily available, accurate and supported out of the box by the Linux kernel.

In order to be able to read out the battery voltage with the ESP32 ADCs, a simple resistor voltage divider circuit was used. With a division ratio of 4:1, on pin 9, IO33, there is 1/5 of the battery voltage. The resistors are in the 100 k $\Omega$  range, ensuring a current in the 10  $\mu$ A range.

To help with debugging, there are many LEDs placed on the communication and power lines along with a few debug connectors.

### 3.8 Design defects

Of course, a project isn't proper if everything works the first time. The PCB has a couple design defects that were able to be bypassed. We will outline some of them here.

**Strapping pins** We made a mistake and overlooked some ESP32 strapping pins. They are pins used by the ESP during boot to set some parameters. The pin we overlooked was GPIO12. This pin sets the voltage level of the internal SPI flash on which the firmware is stored. This pin is shared by the SPI bus used to communicate with the TSS, and as a result the state of the pin was always undetermined. This resulted in the board having issues booting and flashing sometimes. It was bypassed by burning an eFuse to set the voltage level permanently.

**MCP and TSS reset pins** The MCP2551 can be put into a low power state by the Rs pin. We had the idea of controlling the state of that pin using a transistor switch. However, the switch was incorrectly designed and the MCP was always in a low power state. This prevented us from transmitting data to the VAN bus. This was bypassed by permanently putting the MCP into a running state with a 10k $\Omega$  pulldown. This resulted in increased idle power consumption by a few mA.

The same issue was with the reset pin on the TSS463C. However, the TSS is always reset by software on boot, so that was easily bypassed by just removing all components around the reset pin.

**Level shifter** The level shifter used for the VAN RX line connected to the ESP32 used the +5V Ignition rail as the reference for the 5V side. This rail, however, was turned off when the ESP decided the Raspberry Pi shouldn't be powered on. As a result, the ESP would power this down and never receive a VAN frame again, preventing the firmware from powering the rail back on. This was bypassed by cutting the trace and connecting the reference to the +5V Battery rail.

## 4 Gateway module software

The gateway module is the most important part of this project as it interfaces with the car and is responsible for the entire system's power management. The module receives and parses VAN frames, organizes the data and then forwards it to the infotainment computer. It also does the reverse. It can receive data from the infotainment computer and transmit the appropriate VAN frame on the bus. The power management is also an important part of the software, as excessive idle power draw will drain the car's battery fairly quickly.

### 4.1 Framework and approach

For the software, we are using the ESP-IDF framework from Espressif. The framework is based on the FreeRTOS Operating System. Due to the nature of the OS and the software environment, the module software was written with an event-based approach.

We use the following events as a base for the main loops/tasks:

- Receiving VAN frames
- Receiving UART data from the computer
- Car state change
- Timer events
- Main task

Since the ESP32 has two cores, we delegated one core to exclusively handle receiving and parsing VAN frames, since they are very frequent and we don't want to miss any.

### 4.2 Receiving VAN frames

To receive VAN frames we have to manually read and time the RX pin states. Luckily, the ESP32 has a hardware peripheral that does just that: the RMT peripheral. It can monitor GPIO pin states and also measure how long it stayed in that state. Once we set the thresholds for the minimum and maximum signal lengths we can have the peripheral trigger an interrupt every time it sees an **EOF** sequence.

---

```
1  rmt_rx_channel_config_t rx_chan_config = {0};
2
3  ...
4
5  timeslot_interval_ns = 8 * 1000; // At 125 kTS/s
6
7  // EOF is 10 TS long
8  rx_rcv_config.signal_range_max_ns = 10 * timeslot_interval_ns;
9  rx_rcv_config.signal_range_min_ns = timeslot_interval_ns / 2;
10
11 ...
```

---

Listing 1: RMT timing initialization

Once the **EOF** sequence is found, the RMT peripheral driver triggers an interrupt. Inside that interrupt, we put a new event in the VAN Receiver task queue and restart the peripheral. The VAN Receiver

task takes the received RMT data from its queue and parses it. The data is an array of structs whose definition can be found in Listing 2. We reconstruct the VAN frame bit by bit from that array, taking into account the Manchester bits. After checking the checksum, we separate the identifier and data, put it in a new buffer and post it to the queue of the Main task for further processing. The receiving logic was inspired by Peter Pinter’s arduino library [6] and adapted to suit our firmware architecture.

---

```

1  /**
2   * @brief The layout of RMT symbol stored in memory,
3   *       which is decided by the hardware design
4   */
5  typedef union {
6      struct {
7          uint16_t duration0 : 15; /*!< Duration of level0 */
8          uint16_t level0 : 1;     /*!< Level of the first part */
9          uint16_t duration1 : 15; /*!< Duration of level1 */
10         uint16_t level1 : 1;     /*!< Level of the second part */
11     };
12     uint32_t val; /*!< Equivalent unsigned value for the RMT symbol */
13 } rmt_symbol_word_t;

```

---

Listing 2: Definiton of the RMT data symbol in ESP-IDF

### 4.3 Parsing VAN frames

Once the VAN receiver task posts a new event, the Main task then calls the packet parser which checks the identifier and, if it’s recognized, parses the data into global buffers and posts appropriate events to the Main queue depending on the packet being parsed. See listing 3 for an example VAN frame struct. The vast majority of known decoded VAN frames can be found on Peter Pinter’s website [7].

---

```

1  /**
2   * @brief VAN rpm and speed data struct. Iden 0x824
3   *
4   */
5  struct psa_van_rpm_speed_struct
6  {
7      uint16_t rpm;           // RPM in big endian. Divide by 8 to get actual rpm;
8      uint16_t speed;        // Speed in big endian. Divide by 100 to get actual speed;
9      uint16_t distance;     // Distance covered. Increments with each passed decimeter
10     uint8_t packet_changed; // Increments each time information significantly changes
11 } __attribute__((packed));

```

---

Listing 3: Example VAN frame struct.

### 4.4 Sending VAN frames

We use the TSS463C to send VAN frames to the car. The TSS is effectively additional 128 bytes of RAM that we have to manage, since we have to manage all the memory used for sending and receiving data. For now, the TSS is only used to send CD Changer packets so we can emulate it, and to send trip reset requests. The CD Changer messages are immediate reply messages (see 2.2.1), and the trip reset is a normal frame with an ACK request. The CD Changer packets are on a timer and sent every second, otherwise the onboard radio will think the CD Changer is absent and disable it.



The TSS communicates over SPI and is a bit cumbersome to use, so a custom driver was written to make it easier to use. Additionally, there is a separate task that monitors the TSS message states and queues them to be sent.

## 4.5 UART communication

The infotainment computer communicates with the ESP32 over UART. The reason we use UART is its simplicity of use. The ESP mostly sends data to the computer, but can also receive requests from the computer.

The uart code lives in its own task that receives events from other tasks and can also send events to the main task. When the Main loop sees a change to the global data buffers (a VAN frame arrived, battery voltage changed, etc...) it posts an appropriate event to the uart task with the data to send to the computer. The protocol and data structs are custom and inspired by the MSP protocol used in Flight Controller firmware on FPV drones.

## 4.6 Event handling and tasks

The firmware uses several tasks that run as their own loops and wait for events from multiple queues. There are a total of 4 global queues in the firmware, one for each major component:

- Main Queue - Read by the Main task. All other tasks post their events to the Main task queue. The Main task then determines what to do and where to forward that event next.
- TSS Queue - Read by the TSS task. Used to queue packets for transmission by the TSS.
- UART Queue - Read by the UART task. Used to send packets to the Infotainment computer.
- VAN Queue (Unused) - Read by the VAN RMT task. Left for future use.

Additionally, there is an internal queue in the VAN RMT task. This queue is used exclusively between the RMT peripheral's interrupt and the VAN RMT task.

The events are defined as a struct (Listing 4) with the event identifier (Listing 5) and a pointer to some data. The data depends on the event being received.

## 4.7 Handling power states

It is important we manage the system's power consumption. To make that easier, we have defined several 'car states'. See table 3 for a list of those states, and the systems behaviour. One important component mentioned in that table is ESP32's deep sleep mode. When placed in deep sleep, the ESP is practically turned off. The only thing running on the chip is a special section of RAM, the RTC peripheral and, if needed, the ULP coprocessor. For our usecase, we use the same ADC pin we use for battery voltage measurement to wake up the ESP. Due to the voltage ranges of the pin, we can't use simple GPIO wake up, but must use the ULP coprocessor to measure voltage and wake the ESP up when it increases past a certain threshold.

## 4.8 The ULP coprocessor

The ESP32 has a very low power coprocessor onboard. It has a very simple architecture and uses very little power. It is ideal to use for ADC measurement and waking up the ESP. It cannot be programmed directly in C, but has two other ways of writing code. The first one is writing the assembly code in



a separate file and building it into the firmware binary. The other one is using predefined C macros and writing the program in an array inside the C program and loading it that way. We opted to use the second option as it's way simpler to use. See Listing 6 for the ULP code used in deep sleep.

| Description                                  | Computer state                                                 | ESP32 state |
|----------------------------------------------|----------------------------------------------------------------|-------------|
| Car in sleep mode (switched 12V rail is off) | Turned Off (no power)                                          | Deep sleep  |
| Car off and locked, not in sleep mode        | Turned Off                                                     | Running     |
| Car off, not locked, not in sleep mode       | If coming from states above, turned off, otherwise display off | Running     |
| Car off, radio turned on                     | Turned on, display on                                          | Running     |
| Car off, key in ACC                          | Turned on, display on                                          | Running     |
| Car off, key in IGN                          | Turned on, display on                                          | Running     |
| Car cranking                                 | Turned on, display off                                         | Running     |
| Engine running                               | Turned on, display on                                          | Running     |

Tablica 3: Car state table

```

1 struct mive_global_event
2 {
3     enum mive_global_event_t event;
4     void* ev_data;
5 };

```

Listing 4: Event struct definition

---

```

1  enum mive_global_event_t
2  {
3      MIVE_EVENT_NONE = 0,
4      MIVE_EVENT_GLOBAL_SLEEP,
5      MIVE_EVENT_RMT_NEW_VAN_FRAME,
6      MIVE_EVENT_TSS_RESET,
7      MIVE_EVENT_TSS_ACTIVATE,
8      MIVE_EVENT_TSS_IDLE,
9      MIVE_EVENT_TSS_SLEEP,
10     MIVE_EVENT_TSS_WRITE_FRAME,
11     MIVE_EVENT_TSS_MONITOR_CHANNEL,
12
13     MIVE_EVENT_VAN_UPDATE_AUDIO_MENU,
14     MIVE_EVENT_VAN_NEXT_AUDIO_MENU_ITEM,
15     MIVE_EVENT_VAN_CLOSE_AUDIO_MENU,
16
17     MIVE_EVENT_UART_NEW_DATA, // New VAN Data to send
18     MIVE_EVENT_UART_RECEIVE, // Received data from UART
19     MIVE_EVENT_UART_SEND, // Send data to UART
20
21     MIVE_EVENT_TIMER_UART_SEND,
22     MIVE_EVENT_TIMER_1MS,
23     MIVE_EVENT_TIMER_1S,
24
25     MIVE_EVENT_ADC,
26     MIVE_EVENT_STATE_CHANGE,
27     MIVE_EVENT_POWER,
28 };

```

---

Listing 5: List of some events present in the firmware

---

```

1  const ulp_insn_t program[] = {
2      I_MOVI(R0, 0), // Set reg. R0 to initial 0
3      I_MOVI(R2, 0), // Set reg. R2 to initial 0
4      M_LABEL(1), // LABEL 1 - Start of measurement
5      I_ADDI(R0, R0, 1), // Increment cycle counter (reg. R0)
6      I_ADC(R1, PSA_ADC_UNIT, PSA_ADC_CHANNEL), // Read ADC value to R1
7      I_ADDR(R2, R2, R1), // Add ADC value from R1 to R2
8      M_BL(1, 4), // If cycle counter is less than 4, go to LABEL 1
9      I_RSHI(R0, R2, 2), // Divide accumulated ADC value in R2 by 4 and save it to R0
10     M_BGE(2, high_adc_threshhold), // If average ADC value from reg. R0 is higher or equal than high_adc_threshhold, go to LABEL 2
11     // R0 is higher or equal than high_adc_threshhold, go to LABEL 2
12     I_HALT(), // Halt the coprocessor, it will be woken up by the timer
13     M_LABEL(2), // LABEL 2
14     I_WAKE(), // Wake up ESP32
15     I_END(), // Stop ULP program timer
16     I_HALT(), // Halt the coprocessor
17 };
18
19 size_t size = sizeof(program)/sizeof(ulp_insn_t);
20 ESP_ERROR_CHECK(ulp_process_macros_and_load(0, program, &size));
21 ulp_set_wakeup_period(0, 20000); // ULP wakeup timer
22 err = ulp_run(0);

```

---

Listing 6: ULP code that runs in deep sleep.

---

```
1 int ret = xQueueReceive(main_queue, &event, pdMS_TO_TICKS(1000));
2
3 if(ret == pdPASS)
4 {
5     switch (event.event)
6     {
7         case MIVE_EVENT_RMT_NEW_VAN_FRAME:
8             van_packet = (mive_van_packet_t*)event.ev_data;
9
10            ret = psa_parse_van_packet(
11                van_packet->iden,
12                van_packet->packet_size,
13                van_packet->packet,
14                &global_libpsa_buffers);
15
16            ...
17            break;
18        }
19    }
```

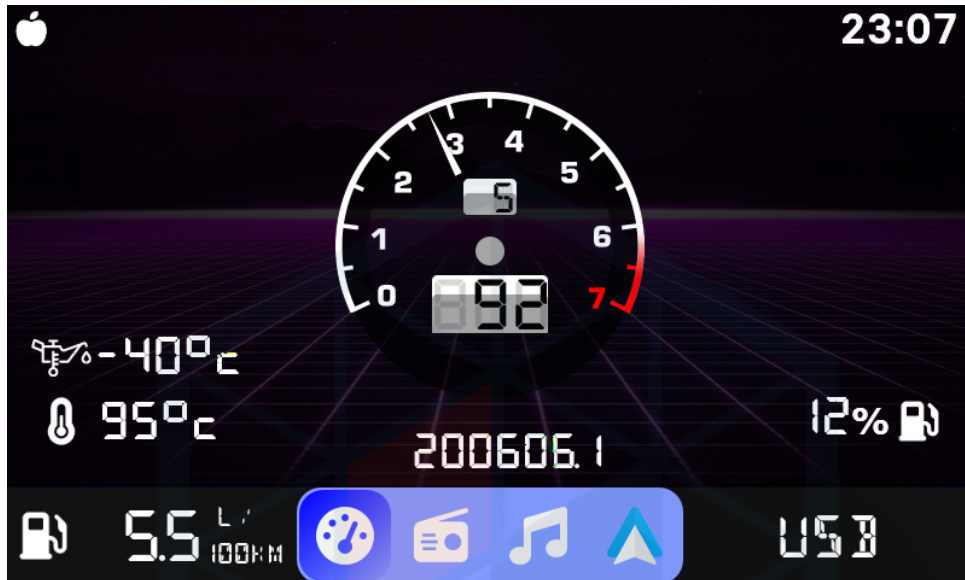
---

Listing 7: Example event handling

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## 5 Infotainment computer software

The infotainment computer is a Raspberry Pi 4 with a DSI touchscreen display. It's running a custom Linux distribution built with the Yocto build system. The graphical application is written using C++ and the Qt framework. This software is not the focus of this assignment and will not be covered in depth.



Slika 14: Screenshot of an early version of the software.

### Features:

- Basic live telemetry. (Engine RPM and temperature, vehicle speed, fuel status)
- Vehicle information. (Door status, trip computer)
- Radio information. (Station information, CD player information)
- Local music player.
- Bluetooth player.
- Time-of-day based brightness control.

## 6 Conclusion and future plans

While designing and working on this project we learned a lot about the inner workings of car ECUs, communication and PCB design. This project provided unique challenges, of which the biggest one was tackling the lesser known and documented (in French no less) VAN protocol. We have already started working on the next revision of the board as well as the next step in this project. Since we are piggybacking on the existing display in the car, the next step is reverse engineering the logic of the display and completely replacing it. That, however, is for another time.

Another challenge that had to be tackled for this project, but wasn't mentioned as much, was building a custom Linux distribution for the Raspberry Pi. That will be expanded upon in the next iteration of this project.



Slika 15: Infotainment system placed in the car

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